

WASP-10b

R-0916

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_180625 · created 2026-07-29T07:55:15+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458294.8998 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.2085 +/- 0.0065
Transit depth (Rp/Rs)^2	4.35 +/- 0.27 [%]
Orbital Inclination (inc)	88.57 +/- 0.65
Airmass coefficient 1 (a1)	1.276 +/- 0.018
Airmass coefficient 2 (a2)	-0.1466 +/- 0.0078
Scatter in the residuals of the lightcurve fit is	1.47 %
Best Comparison Star	None
Optimal Aperture	4.2
Optimal Annulus	8.73
Transit Duration (day)	0.096 +/- 0.0027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.2085 $\pm$ 0.0065 vs 0.1592 (deviation 7.58 σ)
Duration fit vs published	0.096 d vs 0.0946 d (deviation 0.52 σ)
Mid-time O-C	-1.4 min
Depth SNR	32.1
Residual scatter	1.47 %

Light curve

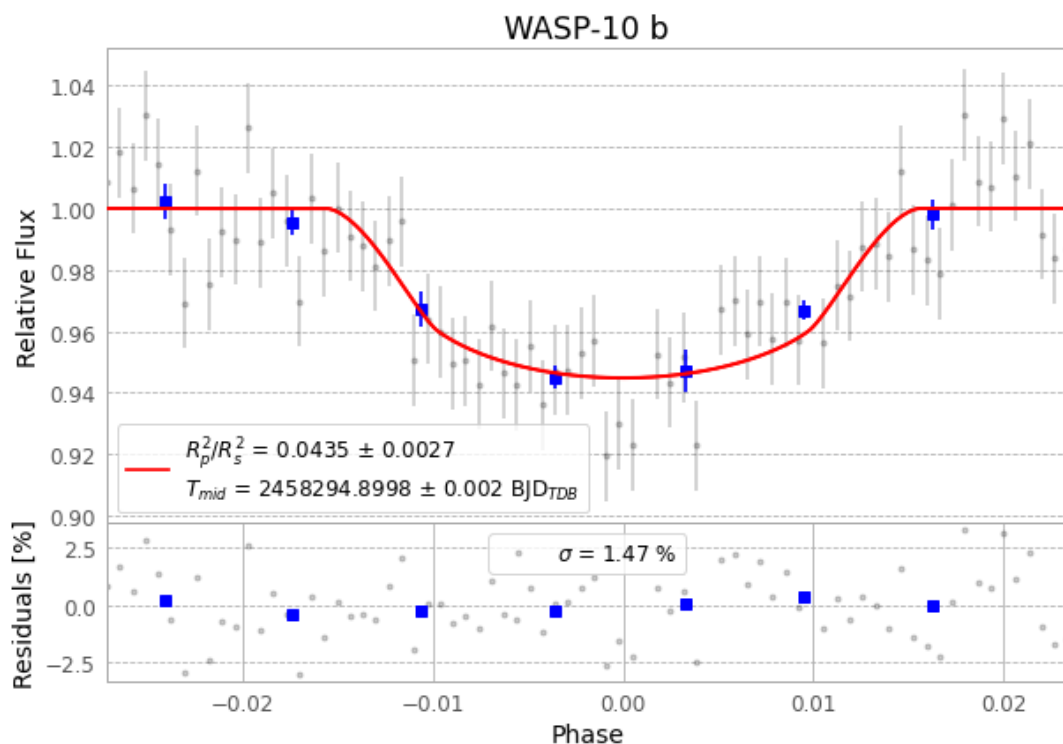


Figure 1 — FinalLightCurve_WASP-10 b_2018-06-25.png (SHA-256 cb74324f6225e1ba...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [372, 152], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b6347b2d4e0b5392...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

74 FITS frames (49 MB), WASP-10180625073312.FITS → WASP-10180625111812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-180625.log (bb488e873fd0...), FinalLightCurve_WASP-10 b_2018-06-25.png (cb74324f6225...), FinalLightCurve_WASP-10 b_2018-06-25.csv (73381674dbc7...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 07:55Z.